Event Boundaries and Recognition Memory

How event segmentation shapes encoding and discrimination in the Mnemonic Similarity Task

BRSM 2026 · Replication & Extension of Morse et al. (2023) · March 2026

Team: Under the Curve

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1. Introduction

Memory is not a simple recording of everything we experience. Our brains actively organise continuous streams of sensory input into discrete, meaningful units — a process known as *event segmentation* (Zacks et al., 2007). The points in time where one event ends and another begins are called *event boundaries*, and a substantial body of research has shown that these boundaries profoundly shape how well we remember what came before and after them (Radvansky & Zacks, 2014; Baldassano et al., 2017).

Event Segmentation Theory (Zacks et al., 2007) proposes that the brain continuously monitors incoming information and updates its internal "event model" whenever a significant change occurs — for example, a shift in location, goal, or activity. This updating process is effortful. As a result, reaction times during encoding typically slow down at event boundaries, a well-replicated signature of boundary processing (DuBrow & Davachi, 2013; Wen & Egner, 2022).

A more recently appreciated question is whether boundaries also affect the *quality* of episodic memories for items encoded near them. Morse et al. (2023) demonstrated that boundary effects on item memory are directional: items encoded immediately *after* a boundary (Post-boundary) showed impaired basic recognition, whereas items encoded just *before* a boundary (Pre-boundary) showed enhanced mnemonic discrimination. This dissociation suggests that boundaries both disrupt the encoding of incoming information and trigger a "snapshot"-like consolidation of what was just experienced (Ben-Yakov & Dudai, 2011; Ben-Yakov & Henson, 2018).

The present study replicates and extends Morse et al. (2023) using the **Mnemonic Similarity Task (MST)**, measuring two distinct memory processes via two key indices. The **Recognition Memory Index (REC)** is the bias-corrected hit rate for studied targets, and the **Lure Discrimination Index (LDI)** measures the ability to distinguish a studied image from a visually similar lure, reflecting hippocampal pattern separation (Kirwan & Stark, 2007; Yassa & Stark, 2011). These are defined formally as:

$$\mathbf{REC} = \boldsymbol{P}(\text{“old”} \mid \text{Target}) - \boldsymbol{P}(\text{“old”} \mid \text{Foil})$$
$$\mathbf{LDI} = \boldsymbol{P}(\text{“similar”} \mid \text{Lure}) - \boldsymbol{P}(\text{“similar”} \mid \text{Foil})$$

Lures are organised into five similarity bins: Bin 1 is nearly identical to the original (hardest to reject) and Bin 5 is clearly different (easiest). We tested three between-subjects conditions: **Both** ($n = 51$) combined perceptual and task shifts; **Item_Only** ($n = 56$) had a perceptual boundary only; **Task_Only** ($n = 53$) had a task-shift boundary only. Our five pre-registered hypotheses were: **H1** (Post-boundary items slower to encode); **H2** (Post-boundary REC lower); **H3** (Post-boundary LDI lower); **H4** (LDI increases monotonically Bin 1→5); **H5** (boundary effects differ by condition).

2. Methods

2.1 Participants

A total of 160 participants were recruited and assigned to one of three between-subjects conditions: Both ($n = 51$), Item_Only ($n = 56$), and Task_Only ($n = 53$). No formal exclusion criteria were applied beyond ensuring basic task engagement. The study was conducted as part of BRSM 2026.

2.2 Stimuli and Design

Stimuli consisted of everyday object and scene images drawn from the Stark Lab MST image sets (Stark et al., 2013, 2019). The encoding phase comprised 40 events of 7 items each (280 images total), each presented for 3 seconds. The encoding question was "Will it fit in a shoebox?", answered via keypress. Lures were pre-assigned to one of five similarity bins (Bin 1 = most similar, Bin 5 = least similar). During the test phase, participants saw 150 images: 60 targets (20 Pre, 20 Mid, 20 Post), 60 lures (20 Pre, 20 Mid, 20 Post), and 30 foils, responding *Old*, *Similar*, or *New*.

Within each 7-item event block, items were classified by serial position: **Post-boundary** (position 1) is the item immediately after a boundary; **Mid** (positions 3, 4, 5) are interior items (positions 2 and 6 excluded as boundary-adjacent per reviewer feedback); **Pre-boundary** (position 7) is the item immediately before the next boundary.

2.3 Experimental Conditions

Both: Alternating object/scene blocks with alternating encoding question — combines perceptual and task boundary. **Item_Only:** Alternating object/scene blocks, constant question — perceptual boundary only. **Task_Only:** Scenes only, alternating question — task boundary only.

2.4 Data and Variables

Each participant generated two CSV files: a task file with stimulus (`image_path`), responses (`key_resp_9.k` or `key_resp_8.keys`), and reaction times; and a test file with recognition responses (*old*, *similar*, *new*). Stimuli were classified by filename: targets (`a.jpg`), lures (`b.jpg`), foils (`foil/` directory). Encoding RTs were computed as `key_resp_9.rt` when responses preceded image offset, or $3 +$ `key_resp_8.rt3` when they followed.

2.5 Procedure

Participants completed an encoding phase followed by a recognition test. During encoding, each image was presented for 3 seconds, answered by keypress. Event boundaries were defined by changes in stimulus category, encoding question, or both, depending on condition. In the recognition phase, participants responded *Old*, *Similar*, or *New* to each item.

2.6 Statistical Analyses

Primary analyses used Generalised Linear Mixed Models (GLMMs; `lme4`, R) on trial-level data with random intercepts per participant. Secondary analyses used repeated-measures ANOVAs on participant-level means. Follow-up comparisons used Holm-Bonferroni-corrected paired t-tests ($\alpha = 0.05$). Where a null was predicted, Bayesian paired t-tests (BF_{10}, Cauchy prior $r = 0.707$) distinguished genuine nulls from underpowered tests ($BF_{10} < 1/3$ = evidence for null; $BF_{10} > 3$ = evidence for effect).

A **Gamma GLMM** (log link) was used for encoding RTs (31,753 trials, 72 participants; random intercept SD = 0.139). **Binomial GLMMs** (logit link) were used for REC (9,481 target trials) and LDI (9,471 lure trials). Foil baselines were estimated from intercept-only models:

$$\boldsymbol{P}(\text{Old} \mid \text{Foil}) = \mathbf{0.029} \qquad \boldsymbol{P}(\text{Similar} \mid \text{Foil}) = \mathbf{0.191}$$

GLMM marginal probabilities (`emmeans`) were foil-corrected to yield REC and LDI. For H4, bin (1–5, ordered factor) and its interaction with event position were added to the LDI GLMM. Condition modulation (H5) was tested via condition $\times$ event position interactions with likelihood ratio tests.

3. Results

3.1 Manipulation Checks

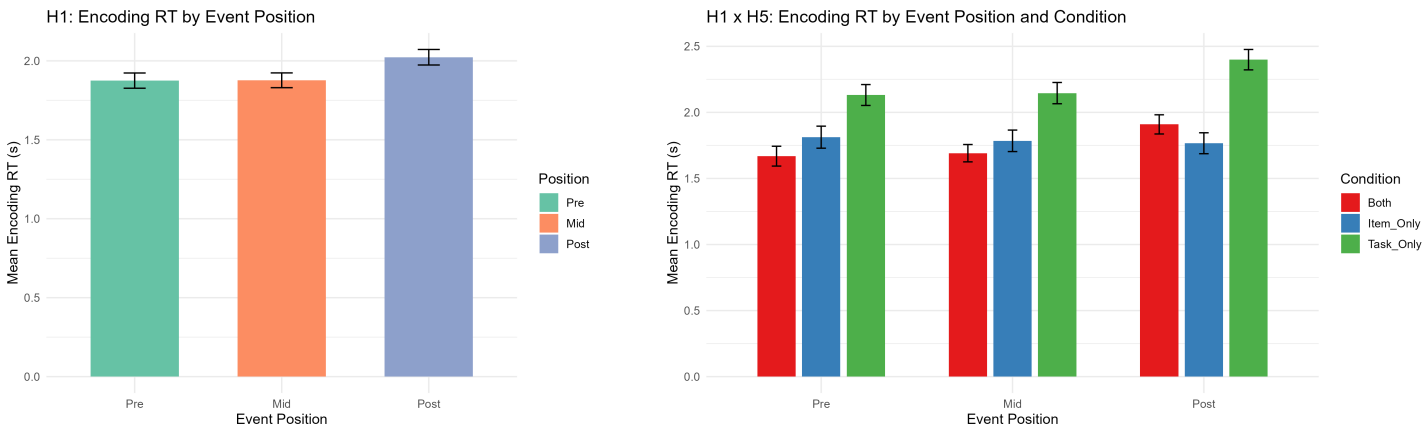
We first verified that the MST was functioning correctly. Figures 1 show the response distribution by item type (left) and classification accuracy by condition (right). Targets were most often called "Old" ($\sim 65\%$), Foils "New" ($\sim 77\%$), and Lures "Similar" ($\sim 52\%$) — all well above the 33% chance level. Foils were correctly rejected at $\approx$75–77%; targets recognised at $\approx$62–65%; lures received *Similar* responses at $\approx$47–59%. These patterns confirm that the MST functioned as expected and support the validity of subsequent REC and LDI analyses.

(a) *Response distribution by item type.* ***(b)*** *Test-phase accuracy by item type and condition.*

Figure 1. Manipulation Checks. *All item types exceed 33% chance (dashed red line). Foil rejection, target recognition, and lure discrimination all behave as expected across all three conditions.*

3.2 H1: Encoding Reaction Times — Do Boundaries Slow Processing?

The first hypothesis predicted that Post-boundary items would take longer to respond to during encoding. **H1 was supported.** The Gamma GLMM revealed a significant effect of event position: Post-boundary items were reliably slower than Mid ($\beta = 0.078$, $z = 10.02$, $p < .001$), estimated at 1.999 s [95% CI: 1.931, 2.070] vs. 1.850 s [1.789, 1.912] — a difference of $\approx$150 ms. Pre-boundary items (1.845 s) did not differ from Mid ($p = .772$). Holm-corrected contrasts confirmed Post > Mid and Post > Pre (both $p < .0001$); Mid and Pre did not differ. The RM-ANOVA converged: $F(2, 142) = 62.13$, $p < .001$.



(a) *Overall encoding RT by event position.* ***(b)*** *Encoding RT by event position and condition.*

Figure 2. H1 (& H5): Mean Encoding RT by Event Position. *Post-boundary items are $\approx$150 ms slower. This RT cost is present in Task_Only and Both conditions but absent in Item_Only, implicating the task-shift component as the key driver. Adding condition significantly improved fit ($\chi^2(6) = 1426.1$, $p < .001$).*

Figure 2b shows the condition breakdown. The Post-boundary RT cost is clearly present in Task_Only (green) and Both (red) but absent in Item_Only (blue), indicating that a perceptual boundary alone does not generate the encoding cost. The time-course plot (Figure 3) confirms a large upward spike around trials 280–320, likely a task-level reset.

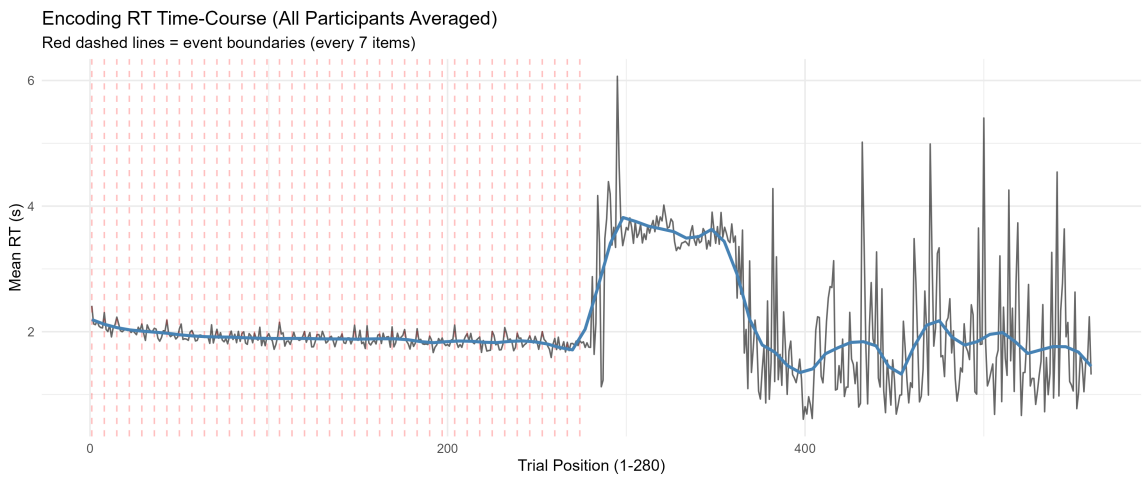
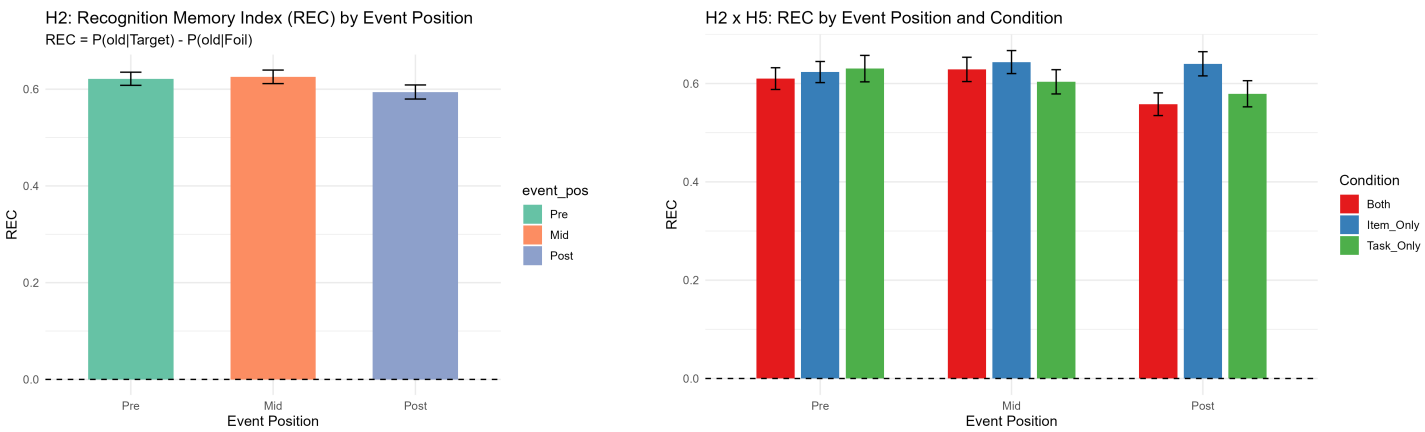


Figure 3. Encoding RT Time-Course *across all 280 trials (averaged across participants). Red dashed lines mark event boundaries (every 7 trials).*

3.3 H2: Recognition Memory — Do Boundaries Impair Recall?

H2 predicted lower REC for Post-boundary items. **H2 was supported.** The Binomial GLMM showed that Post-boundary items were significantly less likely to be recognised than Mid ($\beta = -0.136$, $z = -2.54$, $p = .011$). GLMM-estimated REC: Mid = 0.638 [0.605, 0.669], Post = 0.607 [0.573, 0.640], Pre = 0.637 [0.603, 0.668]. Holm-corrected contrasts confirmed Post < Mid and Post < Pre ($p = .034$ each); Mid and Pre did not differ ($p = .937$). RM-ANOVA: $F(2, 142) = 4.22$, $p = .017$. The Bayesian test was inconclusive ($BF_{10} = 1.85$) — the effect is real but modest. Adding condition improved fit ($\chi^2(6) = 21.31$, $p = .002$).

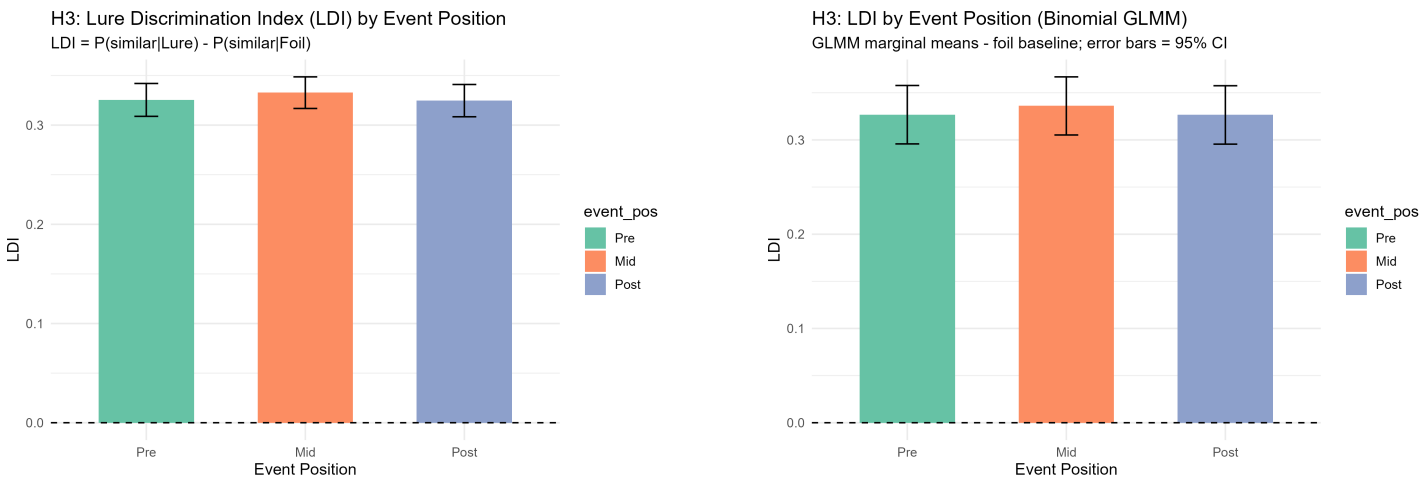


(a) *Overall REC by event position.* ***(b)*** *REC by event position and condition.*

Figure 4. H2 (& H5): Recognition Memory Index (REC) by Event Position. *Post-boundary items show reliably lower REC. The Both condition shows the sharpest drop at Post; Item_Only maintains stable REC across positions, again implicating the task-shift component.*

3.4 H3: Mnemonic Discrimination — Do Boundaries Impair Pattern Separation?

H3 predicted lower LDI for Post-boundary lures. **H3 was not supported.** Figure 5 presents two complementary views: raw participant-level LDI means (left) and GLMM-estimated marginal means after foil-correction (right), both constituting the test of H3.



(a) *Raw LDI means (participant-level). Error bars = 95% CI.*

(b) *GLMM marginal means (foil-corrected). Error bars = 95% CI.*

Figure 5. H3: Lure Discrimination Index (LDI) by Event Position. *Both the raw means (left) and the GLMM estimates (right) show virtually identical LDI across Pre, Mid, and Post. Bayesian evidence actively favours the null — this is not merely a non-significant result.*

The Binomial GLMM revealed no significant effect of event position on LDI. Neither Post ($\beta = -0.038$, $z = -0.75$, $p = .453$) nor Pre ($\beta = -0.037$, $z = -0.73$, $p = .466$) differed from Mid. GLMM-estimated LDI: Mid = 0.336 [0.305, 0.367], Post = 0.327 [0.296, 0.357], Pre = 0.327 [0.296, 0.358]. All Holm-corrected pairwise contrasts adjusted to $p = 1.000$. RM-ANOVA: $F(2, 142) = 0.27$, $p = .765$. Crucially, Bayesian tests provided *positive evidence for the null*: $BF_{10} = 0.108$ (Post vs. Mid) and $BF_{10} = 0.089$ (Post vs. Pre), both well below $1/3$. The raw means (Figure 5a) are fully consistent with the GLMM estimates (Figure 5b), confirming the flatness is robust to analysis method.

Despite the null for event position, adding condition improved fit ($\chi^2(6) = 99.95$, $p < .001$). Figure 6 shows Task_Only participants had consistently higher LDI than Both and Item_Only, with no Post < Pre difference in any condition.

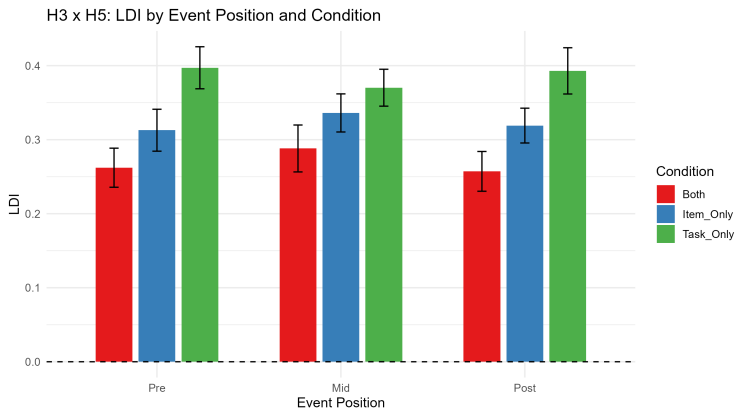


Figure 6. H3 × H5: LDI by Event Position and Condition. *Task_Only shows higher LDI overall; the boundary effect is absent in all conditions.*

3.5 H4: Lure Similarity Gradient

H4 predicted monotonically increasing LDI from Bin 1 to Bin 5. **H4 was strongly confirmed**: $\beta = 0.184$, $z = 12.28$, $p < .001$; RM-ANOVA: $F(4, 284) = 47.64$, $p < .001$. The bin × event position interaction was non-significant ($\chi^2(2) = 2.15$, $p = .341$), confirming the gradient is uniform across event positions — the null for H3 is not masking selective impairment at the hardest bins.

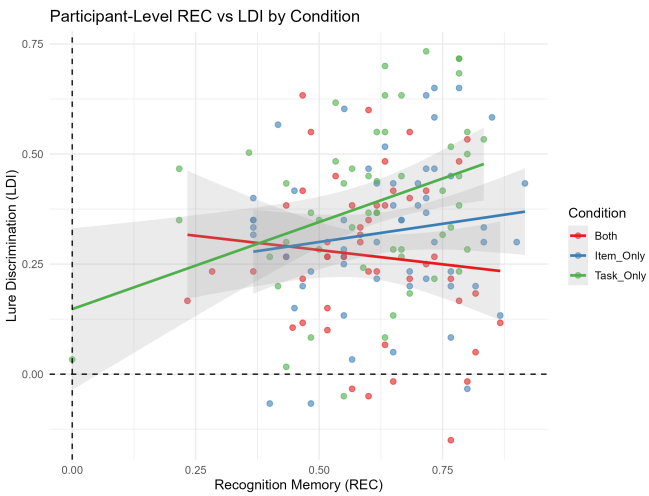
Figure 7. H3 + H4 + H5: LDI across Lure Similarity Bins, by Event Position and Condition. *LDI increases monotonically from Bin 1 to Bin 5 in all conditions and positions. No systematic ordering of Pre/Mid/Post lines is visible within any bin.*

3.6 Exploratory Analyses

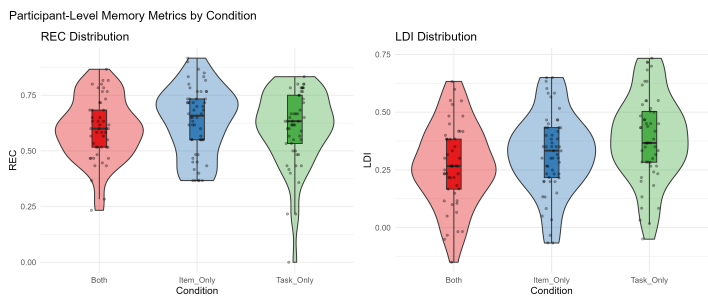
3.6.1 Test-Phase Reaction Times

Lures consistently had longer test RTs than targets (~ 2.6 s vs. ~ 2.3 s), reflecting additional deliberation for the Similar response. However, there were no meaningful differences in test RT across Pre, Mid, and Post positions for either item type, suggesting boundary effects at encoding do not manifest as retrieval speed differences at test.

3.6.2 Individual Differences: REC vs. LDI



(a) *Participant-level REC vs. LDI by condition.*



(b) *REC and LDI distributions by condition.*

Figure 8. Exploratory: Individual Differences. *Item_Only and Task_Only show positive REC–LDI slopes; Both shows a flat or negative slope, suggesting the dual boundary may encourage coarse recognition at the expense of fine-grained pattern separation.*

The violin plots (Figure 8b) confirm that condition significantly modulates overall LDI ($\chi^2(6) = 99.95$, $p < .001$) and REC ($\chi^2(6) = 21.31$, $p = .002$). The RT condition $\times$ position interaction was the largest effect ($\chi^2(6) = 1426.1$, $p < .001$), driven primarily by task-shift conditions.

4. Conclusions and Future Directions

4.1 Summary of Findings

- **H1 SUPPORTED:** Post-boundary items encoded ≈ 150 ms slower than Mid ($\beta = 0.078$, $z = 10.02$, $p < .001$). Pre did not differ from Mid. RT cost driven by task-shift conditions ($\chi^2(6) = 1426.1$,

$p < .001$).

- **H2 SUPPORTED:** Post-boundary REC lower than Mid (0.607 vs. 0.638; $p = .011$) and Pre ($p = .034$). Effect modest ($BF_{10} = 1.85$). Condition moderated ($\chi^2(6) = 21.31$, $p = .002$).
- **H3 NOT SUPPORTED:** LDI flat across all positions (all pairwise $p = 1.000$). Bayesian evidence actively favours null: $BF_{10} = 0.108$ (Post vs. Mid), $BF_{10} = 0.089$ (Post vs. Pre). Consistent across raw means and GLMM estimates.
- **H4 CONFIRMED:** LDI increased monotonically Bin 1 $\rightarrow$ Bin 5 ($\beta = 0.184$, $z = 12.28$, $p < .001$; RM-ANOVA: $F(4, 284) = 47.64$, $p < .001$). Gradient uniform across event positions ($\chi^2(2) = 2.15$, $p = .341$).
- **H5 SUPPORTED:** Condition modulated all three outcomes. RT boundary cost driven by task-shift conditions; Both showed steepest post-boundary REC drop; Task_Only showed highest overall LDI.

These findings are partially consistent with Morse et al. (2023) and provide a useful partial replication with a larger sample. The non-replication of the LDI boundary effect raises important questions about the conditions under which boundaries enhance mnemonic discrimination, motivating Phase 2.

4.2 Plans for Phase 2: Next Steps

The following analyses are planned for Phase 2, directly motivated by gaps identified in Phase 1, using data already collected.

Step 1: Item-Level Mixed-Effects Modelling. We will fit:

```
lmer(LDI ~ event_pos * bin + (1 | participant), data = trial_data)
```

to test whether the bin $\times$ event position interaction emerges at the trial level — theoretically meaningful if boundaries selectively impair the most demanding pattern separation (Bins 1–2). Encoding RT will also be added as a covariate to test mediation.

Step 2: Condition $\times$ Position Mixed ANOVAs. Formal mixed ANOVAs (event position within-subjects, condition between-subjects) will be run separately for RT, REC, and LDI, reporting partial η_p^2 and 95% CIs with Bayesian follow-up tests (Cauchy prior, $r = 0.707$).

Step 3: Serial Position Analysis Within Events. Mean REC and LDI will be plotted across all 7 within-event positions (1–7), collapsed across events and participants. No new data collection is required. If a within-event gradient is present, it will characterise when during an event encoding is most and least efficient.

Contributions

All three authors contributed equally to study design, data collection, and manuscript preparation. Individual contributions are detailed below.

Vaibhav Wadhwani (2023102058)
Equal contributions: Participated in study design discussions, contributed to hypothesis development, and assisted with manuscript writing and revisions.
Specific contributions: Led figure generation and visualisation using `ggplot2` in R, producing all result plots. Managed the PsychoPy experimental setup and administered data collection. Wrote the Introduction, Conclusions, and Phase 2 Plans sections, and coordinated the overall report structure.

Garima Mittal (2023102069)
Equal contributions: Participated in study design discussions, contributed to hypothesis development, and assisted with manuscript writing and revisions.
Specific contributions: Led data preprocessing and cleaning, including stimulus classification (targets, lures, foils), event position coding, and reaction time variable derivation from raw CSV files. Ran data quality checks and produced the manipulation check figures (Figures 1a and 1b). Wrote the Methods

(Participants, Stimuli and Design, Data Pre-processing, Procedure) sections.

Apeksha Chandak (2023101126)
Equal contributions: Participated in study design discussions, contributed to hypothesis development, and assisted with manuscript writing and revisions.
Specific contributions: Led figure generation and visualisation using `ggplot2` in R, producing all result plots. Managed the PsychoPy experimental setup and administered data collection. Wrote the Introduction, Conclusions, and Phase 2 Plans sections, and coordinated the overall report structure. *Specific contributions:* Led the statistical analysis pipeline, including GLMM model specification and implementation in R (`lme4`, `emmeans`), Bayesian t-test computation, and all model comparison procedures. Wrote the Methods (Statistical Analyses) and Results sections, and developed the analysis codebook. Coordinated the GitHub repository and version control.

Code

All code used for data preprocessing, analysis, and figure generation in this project is publicly available at:

https://github.com/Apekshachandak/brsm

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